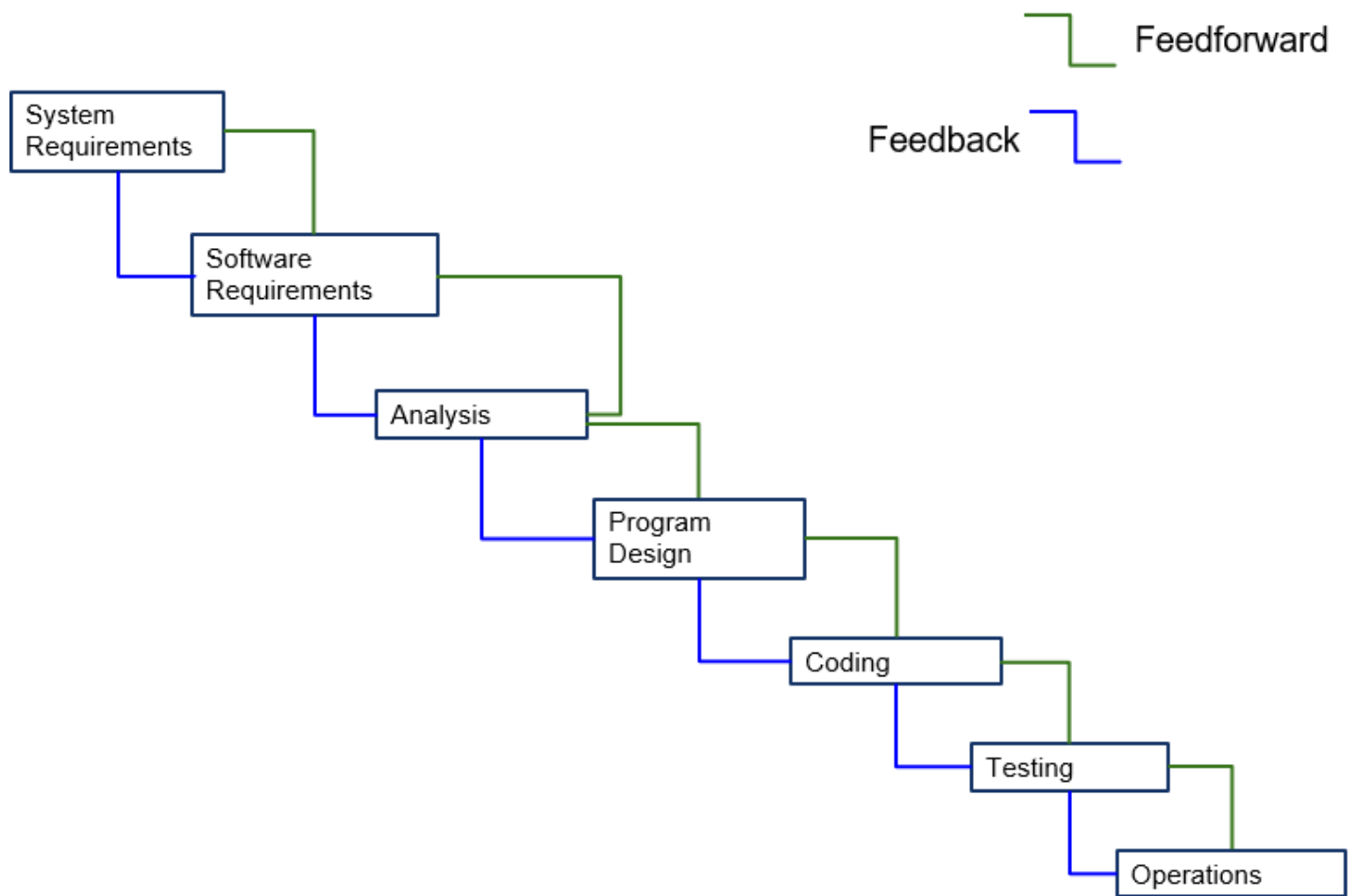


Waterfall Model Architecture – Behavioral & Financial Influence on Credit Score Classification



Box (order)	Process	Team Requirement
1	We will define the main research objective, outline project deliverables, and clarify the scope based on behavioral and financial factors' influence on credit scores. Documentation of essential questions and project metrics will be completed before any technical steps begin.	Project Owner outlines core research goals and deliverables; Peer feedback helps refine initial questions and aligns project scope with expected outcomes.
2	We will specify data requirements (variables, structure), clean the dataset, and establish the protocols for data handling. Key tasks include identifying relevant variables, handling missing values, and forming research hypotheses.	Data Analyst prepares dataset, performs initial descriptive analysis; Project Owner ensures variables match the research focus and documents data-import and cleanup steps.
3	We will conduct exploratory data analysis to identify key patterns or outliers, run statistical tests (e.g., Random Forest, Kruskal-Wallis), and iteratively refine the analytical approach. Results and insights will be documented for transparency.	Analyst runs EDA, builds models, and drafts insights; Peer or advisor may provide review or troubleshooting insights for method selection.
4	We will design the modeling pipeline for data transformation and analysis, including visualization workflows (stacked bars, feature importances, etc.), and plan reporting structure for consistency.	Project Designer develops modeling and reporting pipelines; Reviewer ensures pipeline reproducibility and reporting clarity.
5	We will code the analysis and visualization scripts in Python, R, and Tableau. All project steps, parameters, and results will be logged in version-controlled documentation for future reference or reproducibility.	Coder/Implementer executes technical builds; Peer reviewer checks that procedures deliver intended outputs and scripts are reusable.
6	We will systematically test outputs, validate models, and review visualization products for accuracy. Testing will address all research questions and verify model assumptions or statistical validity.	Quality Controller (solo or peer) validates model results against real outcomes and published findings to ensure reliability.
7	We will document findings, reflect on project limitations, propose next steps, and prepare all deliverables for presentation or broader use. Final project review ensures clarity before handoff.	Project Lead compiles reporting, finalizes documentation, and integrates review notes for improvement and transparency.